

Yeshwanth P

Sr Data Engineer | Email: yesh1th121@gmail.com | contact: 469-214-5151 | Dallas, TX

Professional Summary:

- **8+ years of expertise** in designing, building, and optimizing scalable data pipelines and ETL processes across **cloud platforms such as AWS and Azure**, ensuring data integrity, security, and high availability.
- **Proficient in leading data engineering tools like Apache Airflow, Informatica, and Spark**, with hands-on experience in **Big Data technologies** including **Hadoop, Kafka, Snowflake, and PySpark** for efficient data processing and storage.
- **Strong data visualization skills**, using platforms like **Tableau, Power BI, and AWS QuickSight** to transform complex datasets into **actionable business insights** for decision-making.
- **Deep programming knowledge** in multiple languages, including **Python, Java, and Scala**, allowing the development and deployment of customized data solutions that address specific business needs and optimize performance.
- **Cloud-based data architecture expertise**, working extensively with **AWS services** (such as Redshift, S3, Glue, Lambda, and Kinesis) and **Azure components** (such as Data Factory, Synapse Analytics, and Azure Databricks) to design and maintain large-scale data infrastructures.
- **Led multiple data integration, migration, and warehousing projects**, migrating legacy systems to modern cloud platforms, with experience in **ETL frameworks** like Talend, Azure Data Factory, and AWS Glue.
- **Hands-on experience with real-time data processing** using AWS Kinesis and Kafka for **streaming data**, coupled with robust orchestration and automation using **Apache Airflow**, ensuring seamless data pipeline execution and management.
- **Skilled in building and optimizing data lakes using cloud technologies**, implementing **data governance** frameworks, schema management, and performance tuning for **Snowflake and Redshift** environments.
- **Experienced in working in Agile environments**, collaborating closely with business stakeholders, data scientists, and other engineering teams to define requirements, map out solutions, and deliver on key milestones.
- **Strong focus on continuous improvement** through the implementation of **code quality practices** (SonarQube, automated testing), infrastructure automation (AWS CLI, CloudFormation, Terraform), and **performance optimization strategies** for data pipelines and cloud services.
- **Expert in managing large-scale data warehousing environments**, using advanced SQL skills for **query optimization**, data modeling, and building complex data transformations.
- **Passionate about staying updated with the latest data engineering technologies**, continuously exploring advancements in **machine learning integration**, real-time analytics, and **AI-driven data processing**.
- **Collaborative problem-solver**, working with cross-functional teams to meet business objectives, while ensuring scalability, security, and long-term sustainability of data architectures.

- **Adept at managing end-to-end data projects**, from gathering business requirements to **solution design, implementation, and deployment**, delivering **high-quality, reliable data solutions** that align with business goals.

Technical Skills:

- **AWS Services:** S3, EMR, RDS, EC2, Redshift, Lambda, SNS, SQS, Kinesis, IAM, AMI, Cloud formation
- **Languages:** SQL, PL/SQL, MATLAB, Json, Java, Scala, Python (NumPy, SciPy, Pandas), Shell Scripting.
- **Data Visualization:** Power BI, Tableau, AWS Quick Sight, Informatica, Microsoft Excel, PowerPoint, SAS.
- **Cloud Platforms:** Azure, AWS.
- **Big Data/ Hadoop Components:** HDFS, Hue, MapReduce, PIG, Hive, HCatalog, HBase, Sqoop, Impala, Zookeeper, Flume, Kafka, Yarn, Cloudera Manager, Kerberos, Pyspark, Airflow, Kafka, Snowlake
- **Database:** MySQL, SQL Server, Oracle, Azure SQL Database, AWS Redshift, Cassandra, DB2, Teradata.
- **Spark Components:** Apache Spark, Data Frames, Spark SQL, Spark, YARN, Pair RDDs
- **Data Modeling Tools:** Erwin Data Modeler, ER Studio, Power Designer.
- **Operating Systems:** Windows, Unix.
- **ETL/Data Warehouse Tools:** Informatica, Azure Data Factory, Talend.
- **Reporting and ETL Tools:** Tableau, Power BI. AWS GLUE, SSIS, SSRS

Professional Experience

Client: Nevro Corp

Role: AWS Data Engineer

Date: SEP 2022 – TILL DATE

Responsibilities:

- **Gathered and analyzed business requirements**, collaborated with business users, project leaders, and developers to design data models and propose improvements in processes.
- Conducted **end-to-end architecture and implementation assessment** of various **AWS services** including Amazon EMR, Redshift, S3, Athena, Glue, and Kinesis.
- Utilized **Hive** as the primary query engine of EMR, built external table schemas for processed data, and created **AWS RDS** as a Hive metastore to integrate metadata from 20 EMR clusters, ensuring data retention even if the EMR was terminated.
- Developed a **shell script** to collect and store user logs in **AWS S3** buckets, maintaining a record of all user actions for security purposes to detect cluster termination and safeguard data integrity.
- Implemented **partitioning and bucketing** strategies in the **Apache Hive** database to enhance query retrieval performance.

- Designed and deployed **ETL pipelines** on S3 parquet files in a data lake using **AWS Glue**. Created a **CloudFormation template** in JSON format for content delivery with cross-region replication through **Amazon VPC**.
- Utilized **AWS CodeCommit** Repository for saving programming logics and scripts, enabling replication to new clusters.
- Employed **multi-node Redshift technology** to implement **Columnar Data Storage, Advanced Compression, and Massive Parallel Processing**.
- Migrated to a quality monitoring program from **AWS EC2 to AWS Lambda** and created logical datasets for quality monitoring in the **Snowflake warehouse**.
- Used **AWS Data Pipeline** for extracting, transforming, and loading data from heterogeneous sources.
- Designed, developed, and deployed a convergent mediation platform for data collection and billing processes using **Talend ETL**.
- Managed **Snowflake schemas** and data warehousing, processing batch and streaming data loads using **Snowpipe and Matillion** from a data lake **AWS S3 bucket**.
- Automated the distribution process for groups receiving inventory and sales data, sending activation reports using **Talend and Redshift**.
- Migrated legacy Data Warehouses from **Oracle/SQL Server/Netezza** to a central Data Lake on **AWS cloud/Snowflake** using **Talend and Informatica**.
- Combined **Python and Snowflake's SnowSQL** to write SQL queries against Snowflake.
- Migrated objects from **Netezza to Snowflake**.
- Developed **PySpark frameworks** to transfer data from **DB2 to Amazon S3**.
- Designed data models for **AWS Lambda applications** performing complex analysis and creating analytical reports for end-to-end traceability and lineage of key business elements from **Aurora**.
- Optimized performance of AWS services used by application teams and ensured **code-level application security** for clients using **IAM roles, credentials, and encryption**.
- Employed **SonarQube** for continuous code quality inspection and automatic bug detection.
- Managed **AWS infrastructure** and automation using **CLI and API**.
- Created **AWS Lambda functions** using Python for deployment management and designed, investigated, and implemented public-facing websites on **AWS**, integrating them with other application infrastructures.
- Developed various **AWS Lambda functions and API Gateways** to submit data via API Gateway accessible via Lambda functions.
- Built **CloudFormation templates** for services such as **SNS, SQS, Elasticsearch, DynamoDB, Lambda, EC2, VPC, RDS, S3, IAM, and CloudWatch**, and integrated them with **Service Catalog**.

Environment: Amazon Web Services, Elastic Map Reduce cluster, EC2s, CloudFormation, Amazon S3, Amazon Redshift, Dynamo DB, Cloud Watch, Hive, Scala, Python, HBase, Apache Spark, Spark SQL, Shell Scripting, Tableau, Cloudera, GitHub, Bash Scripting, Bitbucket, CircleCI.

Client: QVC

Role: *AWS Data Engineer*

Date: *July 2020 to Dec 2021*

Responsibilities:

- **Collaborated with Architects and Subject Matter Experts** to review business requirements and build sources to target data mapping documents and pipeline design documents.
- Worked in an **Agile environment** with weekly **Sprints** and daily **Scrum** meetings.
- Utilized **AWS Data Pipeline** for data extraction, transformation, and loading (ETL) from heterogeneous data sources.
- Experienced with **AWS Kinesis** for real-time data streaming and processing, including **Kinesis Data Streams, Kinesis Data Firehose, and Kinesis Data Analytics**. Integrated Kinesis with other AWS services such as **S3, Lambda, and DynamoDB**.
- Familiar with SQL-based data warehousing and reporting tools such as **Tableau and PowerBI**.
- Wrote to the **Glue Metadata Catalog**, enabling querying of refined data from **Athena**.
- Developed **PySpark** jobs in **AWS Glue** to merge data from multiple tables, utilizing Crawlers to populate the Glue Data Catalog with metadata table definitions.
- Designed and developed **Spark workflows** using **Scala** for data pull from **AWS S3 bucket** and **Snowflake**, applying transformations.
- Implemented **Spark RDD transformations** to map business analysis and applied actions on top of transformations.
- Developed PySpark code for **AWS Glue jobs** and **EMR**.
- Automated resulting **scripts** and workflows using **Apache Airflow and shell scripting** to ensure daily execution in **production**.
- Created scripts to read **CSV, JSON, and Parquet files** from **S3 buckets in Python** and load them into **AWS S3, Snowflake DB, and Redshift**.
- Implemented **AWS Lambda** functions to run scripts in response to events in **Snowflake DB tables or S3 buckets or to HTTP** requests using **Amazon API Gateway**.

Environment: HDFS, Map Reduce, AWS S3, Glue, Lambda, Matillion, Snowflake DB, Hive, Sqoop, Pig, Oozie Scheduler, Shell Scripts, Teradata, Oracle, Redshift, HBase, Cassandra, Cloudera, Kafka, Spark, Scala, Python, GitHub, boto, flake8, flask, CircleCI.

Client: DXC Technology

Role: *Data Engineer*

Date: *July 2018 to July 2020*

Responsibilities:

- Developed **Java-based MapReduce** Jobs for efficient data processing with **Hive** and **Pig**.
- Utilized **Zookeeper** for cluster coordination and **workflow** scheduling.
- Automated **UNIX shell scripts, Informatica sessions, and batches** using **Autosys**.
- Proficient in **Informatica PowerCenter** for ETL development and designing **data integration** workflows.
- Implemented **OLAP** multi-dimensional cube functionality using **Azure SQL Data Warehouse/Synapse Analytics**.

- Wrote **Azure PowerShell scripts** for data movement between local file systems and **HDFS Blob storage**.
- Automated Data Processing with **Oozie** to streamline data loading into **Hadoop Distributed File System (HDFS)**.
- Developed custom **Airflow** operators and sensors for specific data engineering requirements.
- Leveraged **Azure Databricks** for big data processing and analytics, setting up clusters, and configuring auto-scaling.
- Designed **end-to-end data pipelines on Azure Databricks**, integrating data from various sources with complex transformations.
- Implemented real-time data processing solutions with **Azure Databricks**, using **Spark Streaming** with **Kafka**.
- Collaborated with cross-functional teams, sharing **Databricks** notebooks and dashboards for **streamlined workflows**.

Environment: Azure, Azure Data Lake, Azure Synapse Analytics, Azure Databricks, Azure Blob Storage, Power BI, PySpark, Scala, Apache Airflow, SQL Data Warehouse, Hive, Zookeeper, Informatica, Oozie, Snowflake, Cassandra.

Client: Info Track Systems

Role: *Jr Data Engineer*

Date: *May 2016 to June 2018*

Responsibilities:

- Developed complex **SSIS/ETL** packages for efficient **data extraction, transformation, and loading**.
- Defined and deployed monitoring, metrics, and logging systems on **AWS** to ensure data system reliability.
- Implemented Data Quality Scripts using **SQL** and **Hive** to validate data load and ensure data quality.
- Created various data visualizations using **Python** and **Tableau** for insightful data representation.
- Utilized **Zookeeper** for storing offsets of messages consumed in **Kafka**, ensuring reliable data processing.
- Developed Entity Relationship Diagrams (ERD), Functional diagrams, and Data flow diagrams using **Erwin**.
- Worked on **AWS cloud** services like **EC2, S3, EMR, Glue, and DynamoDB** for big data processing.
- Designed and maintained **ECS clusters** for high availability and fault tolerance.
- Optimized and tuned **Redshift** environment for faster query performance in **Tableau and SAS Visual Analytics**.
- Designed data pipelines for ingesting and transforming data from data lakes using **EMR**.
- Developed custom EMR applications in **Java, Python, or Scala** to extend EMR's capabilities.

Environment: Hadoop, NoSQL, AWS, AWS Redshift, Apache Kafka, Java, Python, PySpark, SQL Server, SSIS, SSRS, MySQL, Tableau.